

QUESTION 11.1. Solve for x :

1.1.1. $x^2 - 3x = 4$

1.1.2. $x^2 - 3x = 5$

1.1.3. $x^2 - 3x < 4$

1.2. Solve for x and y simultaneously:

$$x - 2y = -1 \quad x^2 + y = xy + 13$$

1.3. For which value(s) of k do the following equations have real roots?

1.3.1. $2^x = k$

1.3.2. $(x+3)^2 = k-1$

1.3.3. $\sqrt{x-1} = k+2$

1.4. For which value(s) of x are the following expressions not real?

1.4.1. $\frac{x}{x-1}$

1.4.2. $\sqrt{4-x}$

1.5. The roots of a quadratic equation are given by $x = \frac{-2 \pm \sqrt{p-5}}{2}$. For which value(s) of p are the roots ...

1.5.1. equal?

1.5.2. non-real?

QUESTION 2 (EXPONENTS REVISION)

2.1. Simplify:

$$2.1.1. \frac{6^x \cdot 3^{x-2}}{6 \cdot 18^{x-2}}$$

$$2.1.2. \frac{2 \cdot 3^x - 3^{x-2}}{2 \cdot 3^{x-2}}$$

$$2.1.3. \frac{{}^4\sqrt{16x^5}}{\sqrt{\sqrt{x}}}$$

$$2.1.4. (\sqrt{2} - 3)(\sqrt{8} + 2)$$

$$2.1.5. \frac{5}{4 - \sqrt{3}}$$

$$2.1.6. \frac{2}{\sqrt{3}} - \frac{\sqrt{3}}{2}$$

$$2.1.7. \frac{2^{2a} - 1}{2^a - 1}$$

2.2. Solve for x :

$$2.2.1. 2^{x+1} \cdot 2^{x+2} = 4$$

$$2.2.2. 2^{x+1} + 2^{x+2} = 6$$

$$2.2.3. 2^{2x} + 3 \cdot 2^x - 4 = 0$$

$$2.2.4. 3^x + 3 \cdot 3^{-x} - 4 = 0$$

$$2.2.5. 4x^{\frac{2}{3}} + 2 = 18$$

$$2.2.6. x^{\frac{2}{3}} + 4x^{\frac{1}{3}} - 5 = 0$$

QUESTION 3

3.1 Solve for x (to 2 decimals if necessary)

$$3.1.1 (x - 1)(2x + 5) = 0 \quad (2)$$

$$3.1.2 \frac{1}{2}x^2 + 3x - 10 = 0 \quad (4)$$

$$3.1.3 x^2 \geq 20 + x \quad (4)$$

$$3.1.4 96 = 3x^{\frac{5}{4}} \quad (3)$$

3.2 Solve for x and y simultaneously:

$$2^{x+y} = 4$$

$$x^2 = 52 - y^2 \quad (7)$$

3.3 Calculate, without using a calculator, the value of a and b if a and b are integers and:

$$\frac{14}{\sqrt{63} - \sqrt{28}} = a\sqrt{b}$$

(4)

QUESTION 4

4.1 Solve for x :

4.1.1 $\frac{x}{3} - \frac{1}{x-1} = 1$ (5)

4.1.2 $2x^2 + 3x - 1 = 0$ (Round off to two decimal places.) (3)

4.1.3 $7x + 4 \leq 2x^2$ (4)

4.2 Solve for m and n simultaneously:

$m^2 - 2mn = 8$ and $m + 3n = -1$ (6)

4.3 Solve for a , without the use of a calculator: $2 \cdot 27^a = \sqrt[3]{72}$ (4)

4.4 Let $Q(x) = \frac{2x-3}{x^2-4}$.

For which value(s) of x is

4.4.1 $Q(x) = 0$? (1)

4.4.2 $Q(x)$ undefined? (2)

[25]

QUESTION 5

5.1 Solve for x :

5.1.1 $x(x+1) = 6$ (3)

5.1.2 $3x^2 - 4x = 8$ (4)

5.1.3 $4x^2 + 1 \geq 5x$ (4)

5.2 Consider the equation: $x^2 + 5xy + 6y^2 = 0$

5.2.1 Calculate the values of the ratio $\frac{x}{y}$. (3)

5.2.2 Hence, calculate the values of x and y if $x + y = 8$. (5)

[19]

QUESTION 6

6.1 Solve for x :

6.1.1 $3x^2 - 7x = 0$ (2)

6.1.2 $5x^2 = 3x + 6$ (4)

6.1.3 $3x^{\frac{2}{3}} - 13x^{\frac{1}{3}} - 10 = 0$ (3)

6.2 Given the expression: $2x^2 - 7x - 15$

6.2.1 Solve for x if $2x^2 - 7x - 15 \geq 0$ (4)

6.2.2 Hence or otherwise, determine for which positive values of x will the following expression be real:

$$\frac{\sqrt{2x^2 - 7x - 15}}{x - 8} \quad (2)$$

QUESTION 7

7.1 Solve for x :

7.1.1 $x^2 - x - 20 = 0$ (2)

7.1.2 $2x^2 - 11x + 7 = 0$ (correct to TWO decimal places) (3)

7.1.3 $5x^2 + 4 > 21x$ (5)

7.1.4 $2^{2x} - 6.2^x = 16$ (4)

7.2 Solve for x and y simultaneously:

$$y + 1 = 2x$$

$$x^2 - xy + y^2 = 7 \quad (6)$$

7.3 The roots of a quadratic equation are given by $x = \frac{-5 \pm \sqrt{20 + 8k}}{6}$, where $k \in \{-3; -2; -1; 0; 1; 2; 3\}$.

7.3.1 Write down TWO values of k for which the roots will be rational. (2)

7.3.2 Write down ONE value of k for which the roots will be non-real. (1)

7.4 Calculate a and b if $\sqrt{\frac{7^{2014} - 7^{2012}}{12}} = a(7^b)$ and a is not a multiple of 7. (4)

Algebra: SolutionsQuestion 1

1.1.1 $x^2 - 3x = 4$

$x^2 - 3x - 4 = 0$

$(x-4)(x+1) = 0$

$x = 4$, OR $x = -1$

1.1.2 $x^2 - 3x - 5 = 0$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(1)(-5)}}{2}$$

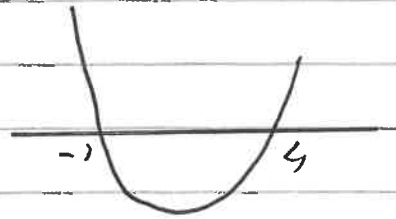
$x = 4, 19$ OR $x = -1, 19$

1.1.3 $x^2 - 3x < 4$

$x^2 - 3x - 4 < 0$

$(x-4)(x+1) < 0$

$-1 < x < 4$



1.2 $x - 2y = -1$ ①

$x^2 + y = xy + 13$ ②

From ① $x = 2y - 1$ into ②

$(2y-1)^2 + y = y(2y-1) + 13$

$4y^2 - 4y + 1 + y = 2y^2 - y + 13$

$2y^2 - 2y - 12 = 0$

$y^2 - y - 6 = 0$

$(y-3)(y+2) = 0$

$y = 3$ OR $y = -2$

$\therefore x = 2(3) - 1$ OR $x = 2(-2) - 1$

$x = 5$

$x = -5$

$$1.3.1. \quad \underline{k \geq 0},$$

$$1.3.2. \quad k - 1 \geq 0 \\ \underline{k \geq 1},$$

$$1.3.2 \quad k + 2 \geq 0 \\ \underline{k \geq -2},$$

$$1.4.1. \quad x - 1 = 0 \\ \underline{x = 1},$$

$$1.4.2. \quad 4 - x < 0 \\ -x < -4 \\ \underline{x > 4},$$

$$1.5.1 \quad p - 5 = 0 \\ \underline{p = 5},$$

$$1.5.2 \quad p - 5 < 0 \\ \underline{p < 5},$$

Question 2

$$2.1.1. \quad \frac{6^x}{6} \cdot \frac{3^{x-2}}{3^{x-2}} \\ = \frac{6 \cdot 18^{x-2}}{(2 \cdot 3)^x \cdot 3^{x-2}} \\ = \frac{2^x \cdot 3^x}{2 \cdot 3 \cdot 2^{x-2} \cdot 3^{2x-4}} \\ = \frac{2^x \cdot 3^{2x-2}}{2^{x-1} \cdot 3^{2x-3}} \\ = \frac{2^{x-x+1} \cdot 3^{2x-2-2x+3}}{1} = 2 \cdot 3 = 6$$

$$\begin{aligned}
 2.1.2. \quad & \frac{2 \cdot 3^x - 3^x \cdot 3^{-2}}{2 \cdot 3^{x-2}} \\
 = & \frac{2 \cdot 3^x - 3^x \cdot 3^{-2}}{2 \cdot 3^x \cdot 3^{-2}} \\
 = & \frac{3^x (2 - 3^{-2})}{3^x \cdot 2 \cdot 3^{-2}} \\
 = & \frac{2 - \frac{1}{9}}{2/9} \\
 = & \frac{17}{2}
 \end{aligned}$$

$$\begin{aligned}
 2.1.3. \quad & \frac{4 \sqrt{16 x^5}}{\sqrt{\sqrt{x}}} \\
 = & \frac{(2^4 x^5)^{1/2}}{\sqrt{x^{1/2}}} \\
 = & \frac{2 x^{5/4}}{x^{1/4}} \\
 = & 2 x^{5/4 - 1/4} \\
 = & 2 x
 \end{aligned}$$

$$\begin{aligned}
 2.1.4 \quad & (\sqrt{2} - 3)(\sqrt{8} + 2) \\
 = & (\sqrt{2} - 3)(2\sqrt{2} + 2) \\
 = & 2(2) - 6\sqrt{2} + 2\sqrt{2} - 6 \\
 = & -4\sqrt{2} - 2
 \end{aligned}$$

$$\begin{aligned}
 2.1.5. \quad & \frac{5}{4 - \sqrt{3}} \times \frac{4 + \sqrt{3}}{4 + \sqrt{3}} \\
 = & \frac{5(4 + \sqrt{3})}{16 - 3} \\
 = & \frac{20 + 5\sqrt{3}}{13}
 \end{aligned}$$

2.1.6.

$$\begin{aligned}
 & \frac{2}{\sqrt{3}} - \frac{\sqrt{3}}{2} \\
 &= \frac{4 - 3}{2\sqrt{3}} \\
 &= \frac{-1}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} \\
 &= -\frac{\sqrt{3}}{6}
 \end{aligned}$$

2.1.7.

$$\begin{aligned}
 & \frac{2^{2a} - 1}{2^a - 1} \\
 &= \frac{(2^a - 1)(2^a + 1)}{(2^a - 1)} \\
 &= \underline{2^a + 1}
 \end{aligned}$$

2.2.1.

$$\begin{aligned}
 2^{x+1} \cdot 2^{x+2} &= 4 \\
 2^{x+1+x+2} &= 2^2 \\
 2^{2x+3} &= 2^2 \\
 2x+3 &= 2 \\
 2x &= -1 \\
 \underline{x} &= \underline{-1/2}
 \end{aligned}$$

2.2.2.

$$\begin{aligned}
 2^{x+1} + 2^{x+2} &= 6 \\
 2^x \cdot 2 + 2^x (2^2) &= 6 \\
 6 \cdot 2^x &= 6 \\
 2^x &= 1 \\
 \underline{x} &= \underline{0}
 \end{aligned}$$

2.2.3

$$2^{2x} + 3 \cdot 2^x - 4 = 0$$

Let $2^x = k$

$$k^2 + 3k - 4 = 0$$

$$(k + 4)(k - 1) = 0$$

$$k = -4 \quad \text{OR} \quad k = 1$$

$$2^x = -4$$

$$2^x = 1$$

No Solution

$$\underline{x = 0}$$

2.2.4

$$3^x + 3 \cdot 3^{-x} - 4 = 0$$

$$3^x + \frac{3}{3^x} - 4 = 0$$

$$3^{2x} - 4 \cdot 3^x + 3 = 0$$

$$(3^x - 3)(3^x - 1) = 0$$

$$3^x = 3$$

$$\text{OR} \quad 3^x = 1$$

$$\underline{x = 1}$$

$$\underline{x = 0}$$

2.2.5

$$4x^{2/3} + 2 = 18$$

$$4x^{2/3} = 16$$

$$x^{2/3} = \frac{16}{4} = 4$$

$$x = 4^{3/2}$$

$$\underline{x = 8}$$

2.2.6

$$x^{2/3} + 4x^{1/3} - 5 = 0$$

$$(x^{1/3} + 5)(x^{1/3} - 1) = 0$$

$$x^{1/3} = -5$$

$$\text{OR} \quad x^{1/3} = 1$$

$$\underline{x = -125}$$

$$\underline{x = 1}$$

Question 3:

3.1.1. $(x-1)(2x+5) = 0$
 $\underline{x = 1}$ or $\underline{x = -5/2}$

3.1.2. $\frac{1}{2}x^2 + 3x - 10 = 0$

$$x = \frac{-3 \pm \sqrt{9 - 4(\frac{1}{2})(-10)}}{2(\frac{1}{2})}$$

$$x = -3 \pm \sqrt{29}$$

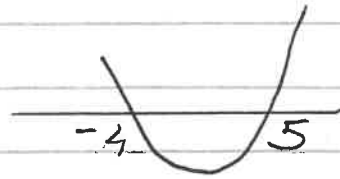
$$\underline{x = -8.39}, \quad \text{or} \quad \underline{x = 2.39}$$

3.1.3. $x^2 \geq 20 + x$

$$x^2 - x - 20 \geq 0$$

$$(x-5)(x+4) \geq 0$$

$$\underline{x \leq -4}, \quad \text{or} \quad \underline{x \geq 5}$$



3.1.4 $3x^{5/4} = 96$
 $x^{5/4} = 32$
 $x = 32^{4/5}$
 $\underline{x = 16}$

3.2. $2^{x+y} = 4$
 $2^{x+y} = 2^2$
 $x+y = 2 \quad \textcircled{1}$
 $x^2 = 52 - y^2 \quad \textcircled{2}$

From ① $y = 2 - x$ into ②

$$x^2 = 52 - (2-x)^2$$

$$x^2 = 52 - (4 - 4x + x^2)$$

$$x^2 = 52 - 4 + 4x - x^2$$

(7)

$$2x^2 - 4x - 48 = 0$$

$$x^2 - 2x - 24 = 0$$

$$(x-6)(x+4) = 0$$

$$\underline{x=6}, \quad \text{OR} \quad \underline{x=-4},$$

$$\therefore \begin{array}{l} y = 2 - 6 \\ \underline{y = -4}, \end{array} \quad \text{OR} \quad \begin{array}{l} y = 2 - (-4) \\ \underline{y = 6}, \end{array}$$

3.3.

$$\frac{14}{\sqrt{63} - \sqrt{28}}$$

$$\frac{14}{\phantom{\sqrt{63} - \sqrt{28}}}$$

$$3\sqrt{7} - 2\sqrt{7}$$

$$= \frac{14}{\sqrt{7}} \times \frac{\sqrt{7}}{\sqrt{7}}$$

$$= \frac{14\sqrt{7}}{7}$$

$$= 2\sqrt{7}$$

$$\therefore \underline{a=2}, \quad \text{and} \quad \underline{b=7}$$

Question 4

$$4.1.1. \quad \frac{x}{3} - \frac{1}{x-1} = 1$$

$$x(x-1) - 3 = 3(x-1)$$

$$x^2 - x - 3 = 3x - 3$$

$$x^2 - 4x = 0$$

$$x(x-4) = 0$$

$$\underline{x=0}, \quad \text{OR} \quad \underline{x=4}$$

4.1.2.

$$2x^2 + 3x - 1 = 0$$

$$x = \frac{-3 \pm \sqrt{9+8}}{4}$$

$$x = -1.78, \quad \text{or } x = 0.28$$

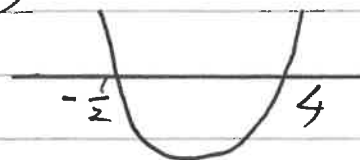
4.1.3

$$7x+4 \leq 2x^2$$

$$0 \leq 2x^2 - 7x - 4$$

$$0 \leq (2x+1)(x-4)$$

$$x \leq -\frac{1}{2}, \quad \text{or } x \geq 4$$



4.2.

$$m + 3n = -1 \quad (1)$$

$$m^2 - 2mn = 8 \quad (2)$$

From (1) $m = -1 - 3n$ into (2)

$$(-1-3n)^2 - 2n(-1-3n) = 8$$

$$1 + 6n + 9n^2 + 2n + 6n^2 = 8$$

$$15n^2 + 8n - 7 = 0$$

$$(15n-7)(n+1) = 0$$

$$n = \frac{7}{15}, \quad \text{or } n = -1$$

$$\therefore m = -1 - 3\left(\frac{7}{15}\right) \quad \text{or} \quad m = -1 - 3(-1)$$

$$= -\frac{12}{5}, \quad = 2$$

4.3.

$$2. \quad 27^a = 3 \sqrt[3]{72}$$

$$2. \quad (3^3)^a = 3 \sqrt[3]{3^2 \cdot 2^3}$$

$$2. \quad 3^{3a} = (3^2 \cdot 2^3)^{1/3}$$

$$2. \quad 3^{3a} = 3^{2/3} \cdot 2$$

$$3^{3a} = 3^{2/3}$$

$$\therefore 3a = \frac{2}{3} \Rightarrow a = \frac{2}{9}$$

$$4.4.1. \quad 2x - 3 = 0$$

$$x = \frac{3}{2}$$

$$4.4.2. \quad x^2 - 4 = 0$$

$$x = 2 \quad \text{OR} \quad x = -2$$

Question 5

$$5.1.1. \quad x(x+1) = 6$$

$$x^2 + x - 6 = 0$$

$$(x+3)(x-2) = 0$$

$$x = -3 \quad \text{OR} \quad x = 2$$

$$5.1.2. \quad 3x^2 - 4x = 8$$

$$3x^2 - 4x - 8 = 0$$

$$x = \frac{4 \pm \sqrt{(-4)^2 - 4(3)(-8)}}{2(3)}$$

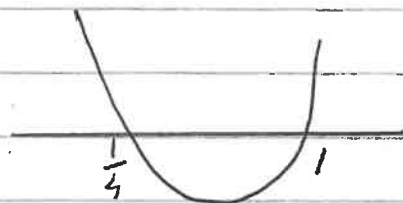
$$x = 2.43, \quad \text{OR} \quad x = -1.09$$

$$5.1.3. \quad 4x^2 + 1 \geq 5x$$

$$4x^2 - 5x + 1 \geq 0$$

$$(4x-1)(x-1) \geq 0$$

$$x \leq \frac{1}{4} \quad \text{OR} \quad x \geq 1$$



$$5.2.1. \quad x^2 + 5xy + 6y^2 = 0$$

$$\frac{x^2}{y^2} + \frac{5xy}{y^2} + 6 = 0$$

$$\left(\frac{x}{y}\right)^2 + 5\left(\frac{x}{y}\right) + 6 = 0$$

$$\left(\frac{x}{y} + 2\right)\left(\frac{x}{y} + 3\right) = 0$$

$$\underline{x = -2}$$

$$y \text{ ————— } \text{,}$$

$$\text{OR } \underline{x = -3}$$

$$y \text{ ————— } \text{,}$$

5.2.2.

$$x = 8 - y$$

$$\therefore \underline{8 - y = -2}$$

OR

$$\underline{8 - y = -3}$$

$$\begin{array}{r} y \\ 8 - y = -2y \\ \underline{y = -8} \end{array}$$

$$\begin{array}{r} y \\ 8 - y = -3y \\ \underline{2y = -8} \\ y = -4 \end{array}$$

$$\therefore x = 8 - (-8)$$

OR

$$x = 8 - (-4)$$

$$\underline{x = 16}$$

$$\underline{x = 12}$$

Question 6

6.1.1.

$$3x^2 - 7x = 0$$

$$x(3x - 7) = 0$$

$$\underline{x = 0}$$

OR

$$\underline{x = \frac{7}{3}}$$

6.1.2.

$$5x^2 = 3x + 6$$

$$5x^2 - 3x - 6 = 0$$

$$x = \frac{3 \pm \sqrt{(-3)^2 - 4(5)(-6)}}{2(5)}$$

$$\underline{x = 1.44}$$

OR

$$\underline{x = -0.84}$$

6.1.3.

$$3x^{\frac{2}{3}} - 13x^{\frac{1}{3}} - 10 = 0$$

$$(3x^{\frac{1}{3}} + 2)(x^{\frac{1}{3}} - 5) = 0$$

$$x^{\frac{1}{3}} = -2^{\frac{3}{3}}$$

OR

$$x^{\frac{1}{3}} = 5$$

$$x = -8$$

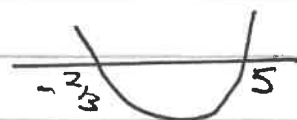
$$\underline{x = 125}$$

$$\underline{27}$$

6.2.1. $2x^2 - 7x - 15 \geq 0$

$$(2x + 3)(x - 5) \geq 0$$

$$\underline{x \leq -\frac{2}{3}}, \text{ OR } \underline{x \geq 5},$$



6.2.2. $\underline{x \leq -\frac{2}{3}}, \quad \underline{x \geq 5}, \quad x \neq 8},$

Question 2:

7.1.1. $x^2 - x - 20 = 0$

$$(x - 5)(x + 4) = 0$$

$$\underline{x = 5}, \text{ OR } \underline{x = -4},$$

7.1.2. $2x^2 - 11x + 7 = 0$

$$x = \frac{11 \pm \sqrt{121 - 4(2)(7)}}{4}$$

$$\underline{x = 4,77}, \text{ OR } \underline{x = 0,73},$$

7.1.3. $5x^2 + 4 \geq 21x$

$$5x^2 - 21x + 4 \geq 0$$

$$(5x - 1)(x - 4) \geq 0$$

$$\underline{x \leq \frac{1}{5}}, \text{ OR } \underline{x \geq 4},$$



7.1.4. $2^{2x} - 6 \cdot 2^x = 16$

$$2^{2x} - 6 \cdot 2^x - 16 = 0$$

$$(2^x - 8)(2^x + 2) = 0$$

$$2^x = 8$$

$$\text{OR } 2^x = -2$$

$$2^x = 2^3$$

No Solution

$$\underline{x = 3}$$

7.2

$$\begin{aligned} y+1 &= 2x & (1) \\ x^2 - xy + y^2 &= 7 & (2) \end{aligned}$$

From (1) $y = 2x - 1$ into (2)

$$\begin{aligned} x^2 - x(2x-1) + (2x-1)^2 &= 7 \\ x^2 - 2x^2 + x + 4x^2 - 4x + 1 &= 7 \\ 3x^2 - 3x - 6 &= 0 \\ x^2 - x - 2 &= 0 \\ (x-2)(x+1) &= 0 \\ \underline{x=2} & \quad \text{or} \quad \underline{x=-1} \end{aligned}$$

$$\therefore y = 2(2) - 1 = \underline{3} \quad \text{or} \quad y = 2(-1) - 1 = \underline{-3}$$

7.3.1. $\underline{k = -2}, \quad \underline{k = 2}$

7.3.2. $\underline{k = -3}$

7.4.

$$\begin{aligned} & \sqrt{\frac{7^{2014} - 7^{2012}}{12}} \\ &= \sqrt{\frac{7^{2012}(7^2 - 1)}{12}} \\ &= \sqrt{\frac{7^{2012} \cdot 48}{12}} \\ &= \sqrt{4 \cdot 7^{2012}} \\ &= 2 \cdot 7^{2012/2} \\ &= \underline{2 \cdot 7^{1006}} \end{aligned}$$